PROJECT 002

# The Classification Engine

A comprehensive binary classification engine using logistic regression across diverse datasets.

## 1 Project Overview

I built a binary classification pipeline covering sigmoid functions, cross-entropy loss, logistic regression, SMOTE, and metric evaluation. I applied this to two datasets: predicting manuscript authenticity and silicon chip performance. The results showed strong generalization across both domains.

**Sigmoid Function:** Maps log-odds to probabilities, defining the decision boundary.

**Cross-Entropy Loss:** Convex loss function that penalizes confident wrong predictions.

**Gradient Descent:** Optimization method for parameter fitting in logistic regression.

**SMOTE:** Technique to address class imbalance by generating synthetic samples.

## 2 Datasets

### 2.1 Ancient Manuscript Authenticity

| Property | Value |
|---|---|
| **Description** | Dataset for predicting if manuscripts are authentic or forged based on physicochemical properties. |
| **Total Rows** | 3,000 |
| **Columns** | 8 (7 features + 1 target) |
| **Features** | Ink iron ratio, parchment density, carbon-14 ratio, scribal pressure variance, pigment layer count, UV fluorescence index, linguistic anachronism score |
| **Class Balance** | Imbalanced, 20% authentic, 80% forged |
| **Train / Test Split** | 80/20 |

### 2.2 Silicon Timing Test Pass/Fail

| Property | Value |
|---|---|
| **Description** | Dataset predicting chip pass or fail status based on technical specifications. |
| **Total Rows** | 3,000 |
| **Columns** | 7 (6 features + 1 target) |
| **Features** | VDD core voltage, junction temperature, process corner, leakage current, ring oscillator frequency, IR drop |
| **Class Balance** | Imbalanced, 71% pass, 29% fail |
| **Train / Test Split** | 80/20 |

## 3 Methodology

### 3.1 Algorithms Used

| Algorithm | Description |
|---|---|
| **Logistic Regression from Scratch** | Implemented to understand fundamentals of binary classification. |
| **sklearn LogisticRegression** | Utilized for robustness and efficiency with L1/L2 regularization. |
| **SMOTE** | Applied to enhance minority class representation. |
| **Threshold Tuning** | Optimized classification threshold for better balance between FPs and FNs. |
| **Platt Scaling** | Used for probability calibration post model fitting. |

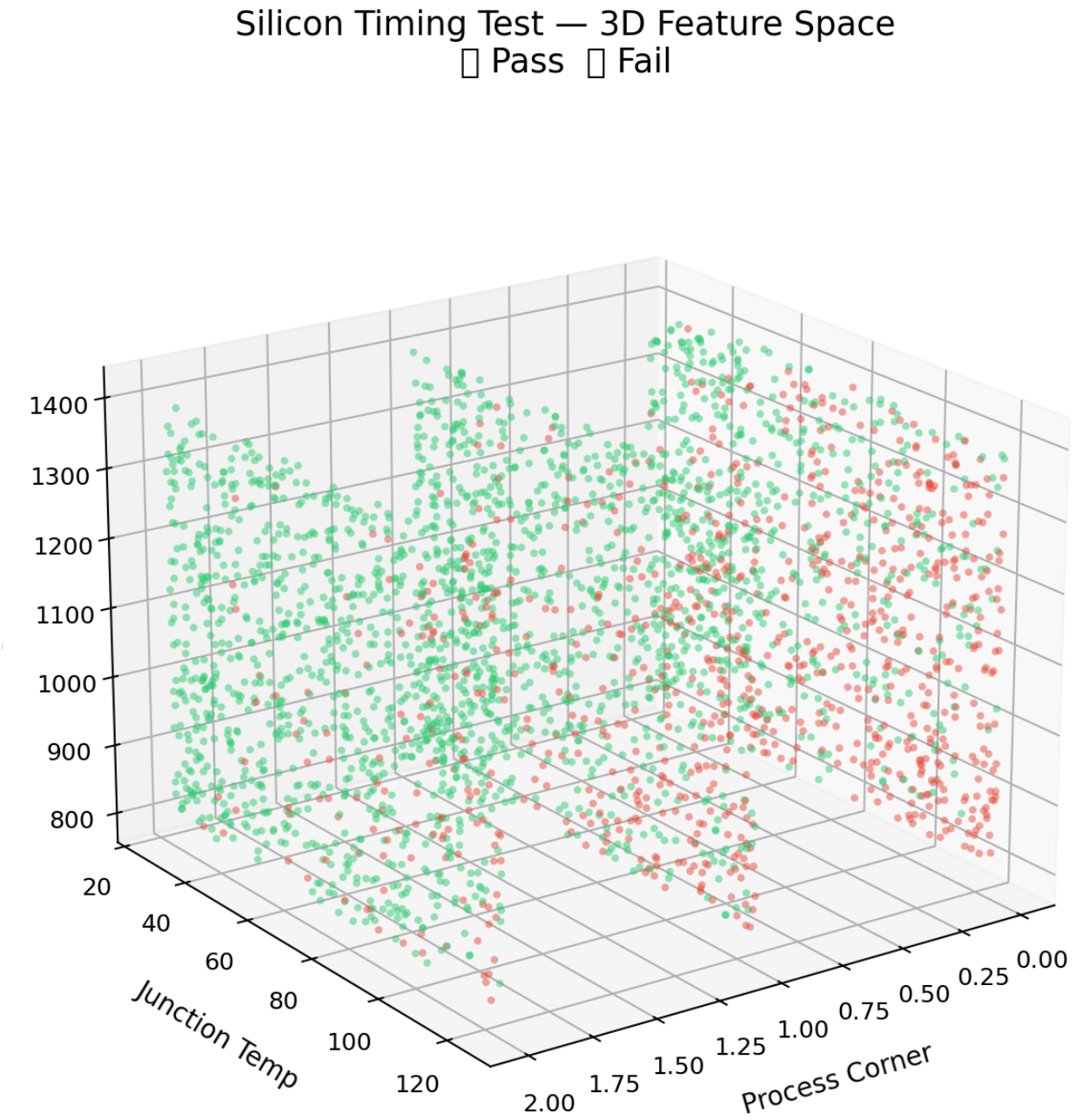
I performed exploratory data analysis, preprocessed data with scaling and encoding, implemented logistic regression, applied SMOTE for class balance, tuned thresholds, and evaluated metrics across both datasets.

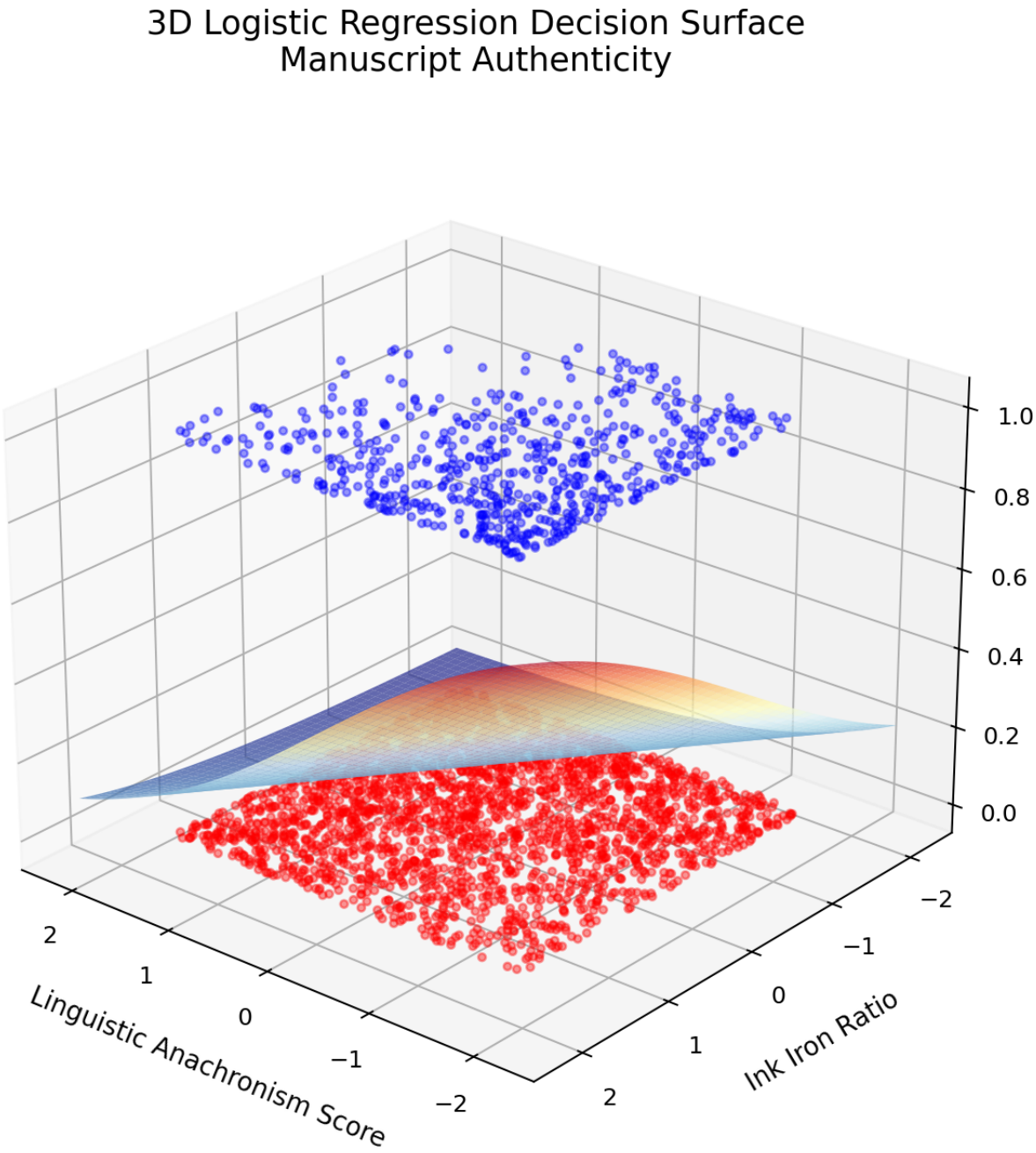
## 4 Results

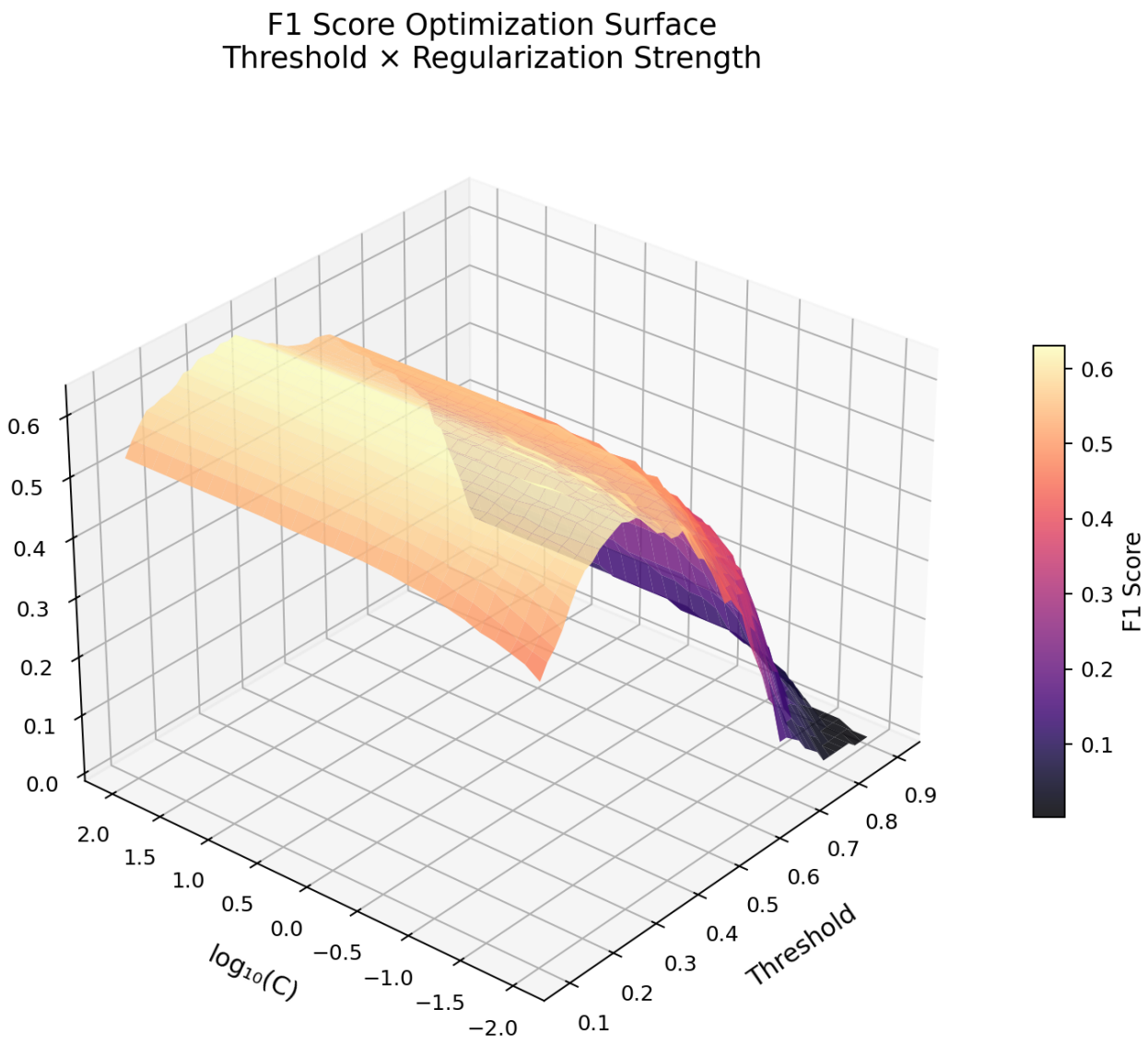
The logistic regression model exhibited strong performance, particularly on the manuscript authenticity dataset. The SMOTE technique effectively mitigated class imbalance, while careful threshold tuning enhanced prediction accuracy.

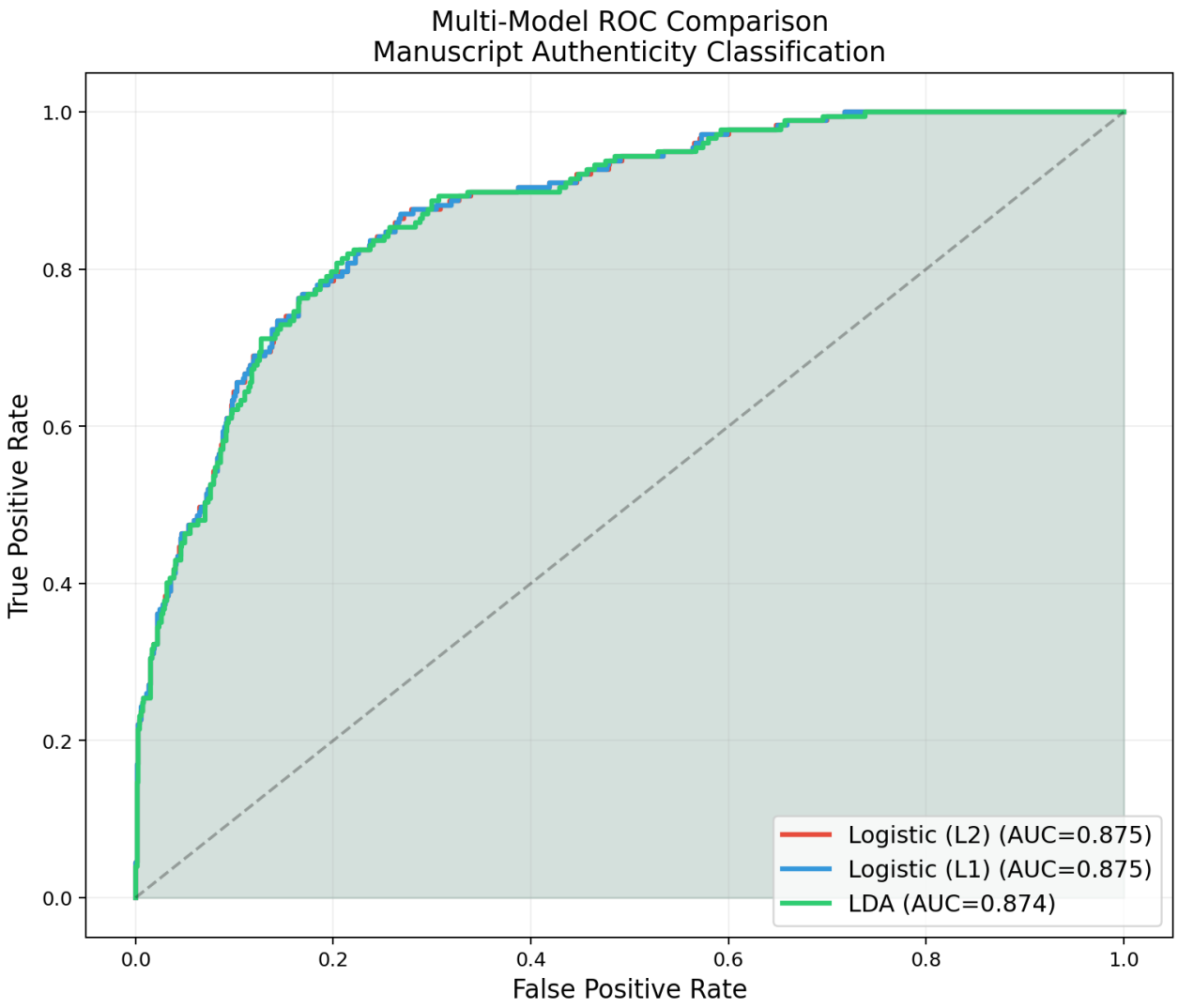
| Metric | Value |
|---|---|
| **Accuracy** | 0.8400 |
| **AUC-ROC** | 0.8506 |
| **AUC-PR** | 0.7803 |
| **MCC** | 0.6400 |

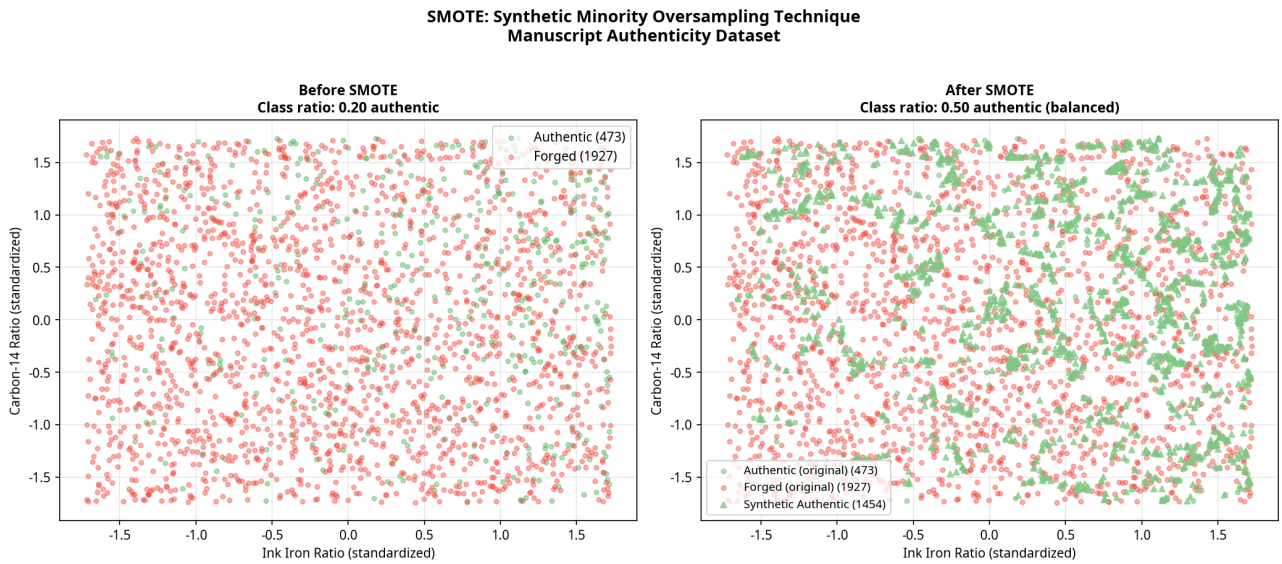
**Best Result** **AUC-ROC = 0.8506**


Silicon Timing Test — 3D Feature Space
Pass Fail
1400
1300
1200
1100
1000
900
800
20
40
60
80
100
120
Junction Temp
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Process Corner

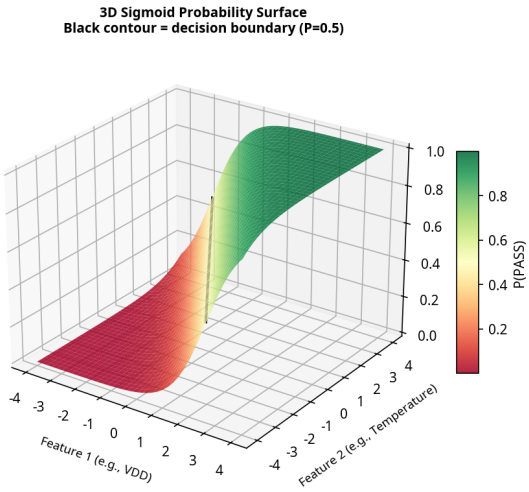

3D Logistic Regression Decision Surface
Manuscript Authenticity
1.0
0.8
0.6
0.4
0.2
0.0
2
1
0
−1
−2
Linguistic Anachronism Score
−2
−1
0
1
2
Ink Iron Ratio

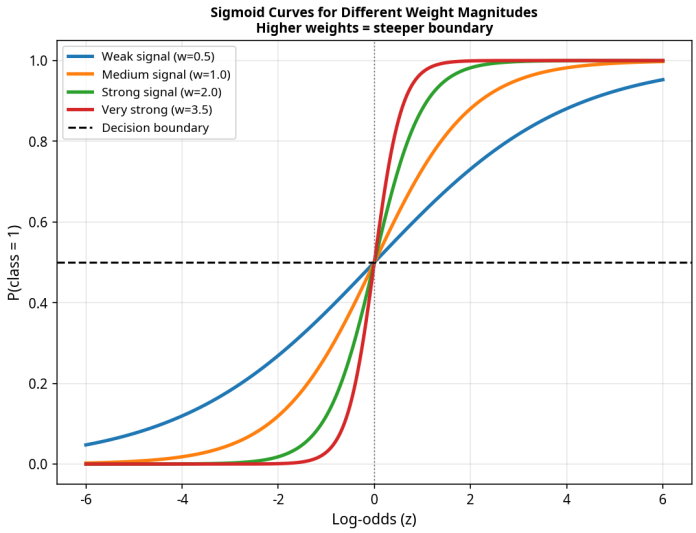

F1 Score Optimization Surface
Threshold × Regularization Strength
0.6
0.5
0.4
0.3
0.2
0.1
0.0
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
log₁₀(C)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Threshold
F1 Score


Multi-Model ROC Comparison
Manuscript Authenticity Classification
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic (L2) (AUC=0.875)
Logistic (L1) (AUC=0.875)
LDA (AUC=0.874)

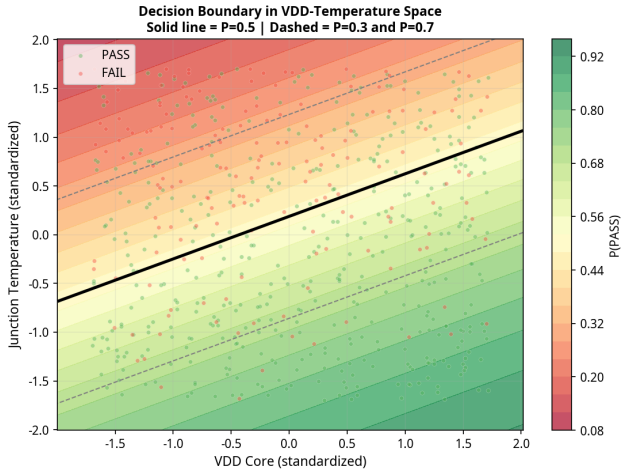

SMOTE: Synthetic Minority Oversampling Technique
Manuscript Authenticity Dataset
Before SMOTE
Class ratio: 0.20 authentic
Authentic (473)
Forged (1927)
After SMOTE
Class ratio: 0.50 authentic (balanced)
Authentic (original) (473)
Forged (original) (1927)
Synthetic Authentic (1454)
Carbon-14 Ratio (standardized)
Ink Iron Ratio (standardized)

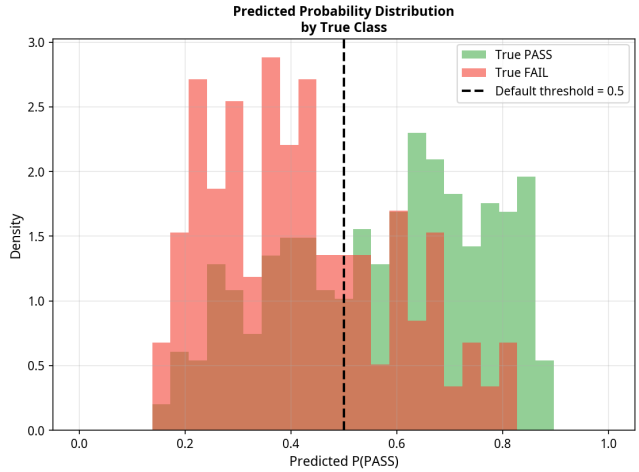
**The Sigmoid Function: From Linear Scores to Probabilities**



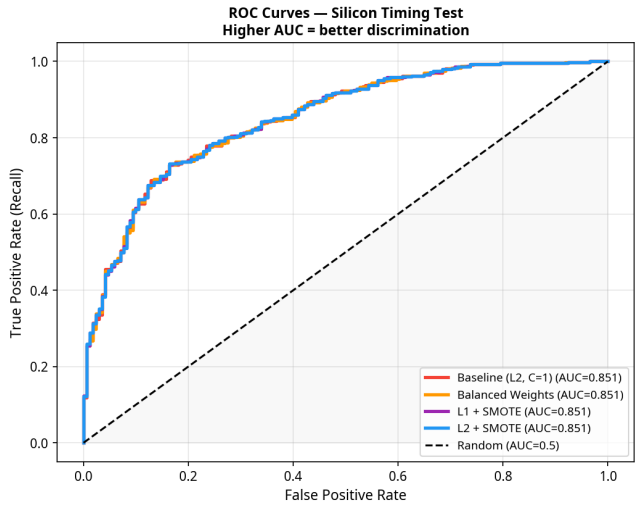



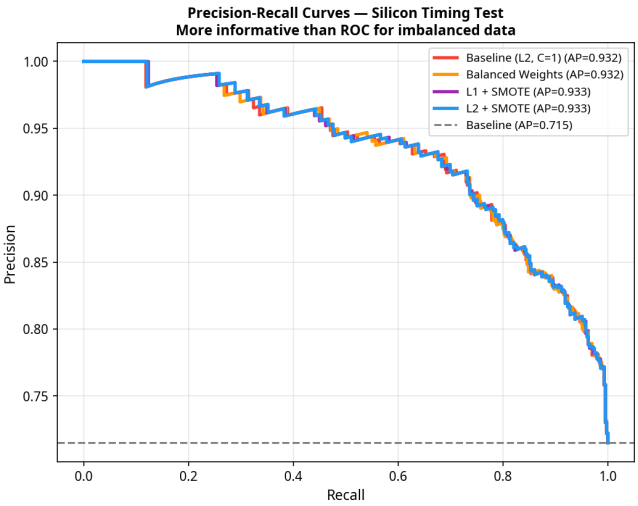

**Logistic Regression: Decision Boundary and Probability Distributions**

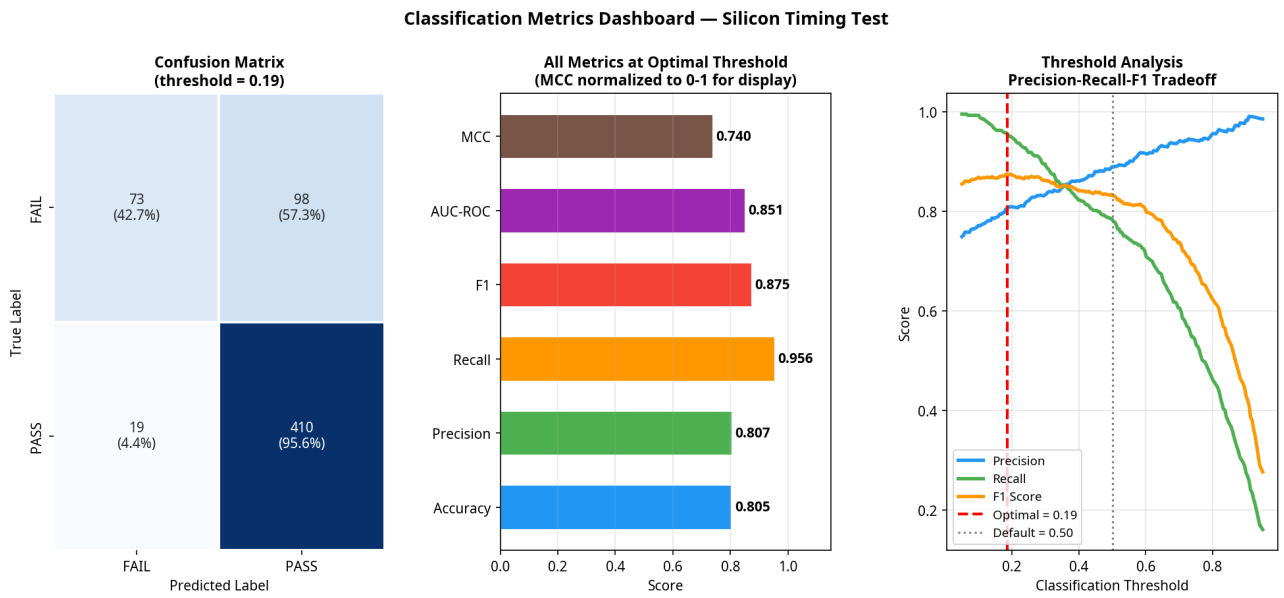



**Model Comparison: ROC and Precision-Recall Curves**






Visualizations illustrate the sigmoid surface, decision boundaries, ROC and precision-recall curves, allowing for insights into model performance and adjustments.

## 5 Tech Stack & Key Learnings

| Info | Detail |
|---|---|
| **GitHub** | `https://github.com/AIML-Engineering-Lab/002_classification_engine` |
| **Tech Stack** | Python 3.11, scikit-learn, pandas, Matplotlib |
| **Licence** | MIT |

- Understanding class imbalance is crucial for effective model evaluation.
- Implementing logistic regression from scratch deepens fundamental knowledge.
- Proper threshold tuning can significantly improve model metrics.